

Arnav Pundir

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EDUCATION

COER University

Roorkee, India

Bachelor of Technology in Computer Science and Engineering

Expected Graduation: 2028

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Artificial Intelligence, Operating Systems, Computer Networks.

SKILLS

Languages : C/C++, Python, JavaScript, TypeScript, SQL, HTML/CSS

Frontend : React.js, Next.js, Tailwind CSS, Three.js, GSAP, Framer Motion

Backend & Systems : Node.js, Express.js, FastAPI, WebRTC, Redis, Distributed Architecture, REST APIs

AI & Tools : OpenCV, YOLOv8, InsightFace ArcFace, PyTorch, Supabase (pgvector), Docker, Git

PROFESSIONAL EXPERIENCE

Freelance Software Developer

Jan 2024 – Present

Independent Projects & Client Deployments

Roorkee, India

- Engineered scalable, high-performance web applications (e.g., SevaSetu) using **React.js** and robust REST APIs, evaluating architectural trade-offs to choose optimal solutions for client requirements.
- Architected interactive WebGL applications utilizing **Three.js**, critically analyzing system performance bottlenecks to consistently achieve **95+ Lighthouse accessibility and speed scores**.

TECHNICAL PROJECTS

TraceAI AI-Powered Intelligence Platform | [GitHub](#)

- Developed an **AI-powered** forensic surveillance platform orchestrating **YOLOv8n**, **ArcFace**, and **ByteTrack** to accurately track and re-identify individuals across video sources.
- Engineered spatial-temporal movement timeline reconstruction and dwell-time analytics across camera zones.
- Architected a **FastAPI** asynchronous backend pushing real-time watchlist hit notifications via **WebSockets**.

AuraMeet Scalable Real-Time Video Chat | [Live](#) | [GitHub](#)

- Architected a highly concurrent, peer-to-peer real-time communication system utilizing **Node.js** and **WebRTC**, optimizing network synchronization for **sub-100ms** global latency.
- Engineered a distributed ephemeral data lifecycle using **Redis pub/sub** to securely manage application state, handling multi-threaded event logic to instantly purge connection data upon user disconnect.
- Programmed a dynamic, responsive video canvas layout using CSS flexbox grids to automatically optimize screen space across devices.

Occasions High-Performance Registration System | [Live](#) | [GitHub](#)

- Developed scalable engineering solutions for a high-traffic event platform using **Next.js 15**, successfully orchestrating concurrent user traffic for **500+ attendees**.
- Applied foundational knowledge in **data structures and algorithms** to design performant debounced search pipelines, substantially reducing database query latency by **40%**.

BioSecure AI Facial Recognition & Template Aging | [GitHub](#)

- Deployed **InsightFace ArcFace 512D** facial embeddings with **Supabase pgvector** similarity queries for multi-face recognition, achieving **90%+** detection accuracy.
- Engineered a 3D Pose-Gated EWMA Embedding Drift Engine ($\alpha = 0.30$) to monitor template aging within a production **Flask** web application.

OPEN SOURCE CONTRIBUTIONS

Core Contributor

Apr 2026 – Jun 2026

Nexus Spring of Code (NSoC) & GirlScript Summer of Code (GSSoC)

Global Rank #70

- Fostered a culture of productivity and innovation by collaborating across globally distributed engineering teams to merge **13 Pull Requests** across 5 core repositories, achieving **Global Rank 70**.
- Developed scalable backend security solutions, actively analyzing software design to mitigate vulnerabilities by implementing rotating **HttpOnly cookie sessions** and tightening CORS policies.